

Primer of Ecology using R

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Preface

UPDATED 2023-08-10

Chapter numbers have changed, starting after Lotka-Volterra competition.

Updated original, but long absent, material on the competition-colonization tradeoff, extinction debt (sensu Tilman 1994), and especially the **storage effect**.

Later this fall, I may add a section on *modern coexistence theory*.

August 2022

These updates will include usual fixes for typos and incremental movement toward slightly practical application. I am trying to include slightly more material and code that students might actually use in their theses and dissertations. When I update the book, I will list below as many of the changes as I can, including

- cleaning up and clarifying the integral projection example.
- includes a simple discrete nonlinear density-dependent population growth model (Watkinson et al. 1980). Several studies evaluating modern coexistence theory and the invasion criterion of coexistence use extensions of this model.
- I will soon add nonlinear density-dependent population growth model (Watkinson et al. 1980) into two species competition.

20 August 2021

NEW IN THIS EDITION OF THE E-BOOK

- Lotka-Volterra versions of direct competition AND mutualism are together in one chapter, followed by consumer-resource interactions, including predator-prey, and resource-based competition and mutualism.
- a chapter on consumer-resource competition and mutualism that follows chapters covering two-species consumer resource interactions (mostly predator-prey).

- If it has been > 1 y since looking at the **primer** package, you might want to peruse the index - I added a few more functions including ones for $1/f$ noise, spectral mimicry, and the Allee effect.

This is based on my book published with Springer, “A Primer of Ecology With R”, part of the Use R! series.

The **primer** package on CRAN (v. 1.2.0) should suffice for this edition. You may also install it directly from GitHub. To do this, install the **devtools** package. To successfully do that on a Windows operating system, I think you need to install Rtools:

<https://cran.r-project.org/bin/windows/Rtools/>

Once you have done all that, use this code:

```
devtools::install_github("HankStevens/primer")
```

In spite of the presumptuous title, my goals for this book are modest. I wrote it as

- the manual I wish I had in graduate school, and
- a primer for our graduate course in Population and Community Ecology at Miami University.¹

Acknowledgements

I am grateful for the generosity of early reviewers and readers, each of whom has contributed much to the quality of this work: Jeremy Ash, Tom Crist, David Gorchov, Raphael Herrera-Herrera, Thomas Petzoldt, James Vonesh, as well as several anonymous reviewers, and the students of our Population and Community Ecology class over the past 15 years. I am also grateful for the many conversations and emails shared with four wonderful mathematicians and theoreticians: Jayanth Banavar, Ben Bolker, Stephen Ellner, and Steve Wright — I never have a conversation with these people without learning something. I have been particularly fortunate to have team-taught Population and Community Ecology at Miami University with two wonderful scientists and educators, David Gorchov and Thomas Crist. Only with this experience, of working closely with these colleagues, have I been able to attempt this book. It should go without saying, but I will emphasize, that the mistakes in this book are mine, and there would be many more but for the sharp eyes and insightful minds of many other people.

¹Miami University is located in the Miami River valley in Oxford, Ohio, USA; the region is home to the Myaamia tribe that dwelled here prior to European occupation.

More recently, Dr. Quan-Guo Zhang has kindly provided very helpful comments and corrections.

I am also deeply indebted to the R Core Development Team for creating, maintaining and pushing forward the R programming language and environment. Like the air I breathe, I cannot imagine my (professional) life without it. I am grateful to all of the developers of packages that have allowed literate programming with Rmarkdown and LaTeX, using RStudio and GitHub.

I am indebted to Rachel Collins for her size-structured coneflower data, to Scott Meiners and his colleagues for their generous sharing of data, meta-data, and statistical summaries from the Buell-Small Succession Study (<http://www.ecostudies.org/bss/>). I would like to thank Stephen Ellner for Ross's Bombay death data and for R code and insight over the past few years. I am also indebted to Tom Crist and his colleagues for sharing some of their moth data (work supported by The Nature Conservancy Ecosystem Research Program NSF DEB-0235369).

Last, and most importantly, I would like to thank those of you in my life who make living worthwhile.

Martin Henry Hoffman Stevens

Oxford, OH, USA, Earth

Today, Current Era

Chapter 1

Theory in Ecology

Placeholder

1.1 Examples of theories

1.1.1 Hierarchy theory

1.1.2 A general theory of ecology

1.1.3 Efficient theory

1.2 An example: Metabolic Theory of Ecology

1.2.1 Body-size dependence

1.2.2 Temperature dependence

1.3 Power law scaling implies constant relative differences

1.4 Meet METE: Maximum Entropy Theory of Ecology

1.5 *In fine*

Chapter 2

Optimal Foraging

Placeholder

2.1 A prey model

2.1.1 Our intuition

2.1.2 Mathematical support

2.2 The patch model

2.3 A simulation of a prey model

2.3.1 Lab exercise

Chapter 3

Simple density-independent growth

Placeholder

- 3.1 Discrete growth rates of fruit flies in my kitchen
- 3.2 Fruit flies with continuous overlapping generations
- 3.3 Properties of geometric and exponential growth
 - 3.3.1 Average growth rate
- 3.4 Modeling with Data: Simulated Dynamics
 - 3.4.1 Data-based approaches
 - 3.4.2 Creating and visualizing the data
 - 3.4.3 One simulation
 - 3.4.4 Multiple simulations
 - 3.4.5 A distribution of possible futures
 - 3.4.6 Analyzing results
 - 3.4.7 Extinction probability: expectation and uncertainty
 - 3.4.8 Inferring processes underlying growth rate
- 3.5 $1/f$ environmental noise
 - 3.5.1 Reddened Song Sparrow growth
- 3.6 *In fine*

Chapter 4

Density-independent Demography

Placeholder

4.1 A two stage matrix model

4.2 A brief primer on matrices

4.3 Decomposing A

4.4 A three stage model

4.5 Projection

4.6 Analyzing the transition matrix

4.6.1 Eigenanalysis

4.6.2 Finite rate of increase

4.6.3 Stable stage distribution

4.6.4 Calculating the stable stage distribution

4.6.5 Reproductive value

4.6.6 Sensitivity and elasticity

4.7 Matrix models can be challenging

4.8 Integral projection

4.8.1 On probability density

4.8.2 A size-based IPM of smooth coneflower

4.8.2.1 Graphing size-dependent transitions

4.8.3 Statistical modeling of size-dependent transitions

4.8.4 Generating the projection kernel

4.8.5 Population summaries

4.9 R packages for demography

4.10 Loggerhead Sea Turtle: a classic example

Chapter 5

Density-dependent growth

Placeholder

5.1 Continuous logistic growth

5.2 Dynamics around the equilibria

5.2.1 A marble and a wok

5.2.2 Analytical linear stability analysis

5.2.3 Projection with numerical integration

5.2.4 Effects of N on birth and death rates

5.2.5 Theta-logistic growth

5.2.6 Allee effect

5.2.7 The integral of logistic growth

5.3 Discrete Density-dependent Growth

5.3.1 Effects of r_d

5.3.2 Bifurcations

5.3.2.1 Sensitivity to initial conditions

5.3.2.2 Boundedness, and other descriptors

5.3.3 The Ricker model

5.4 Nonlinear Discrete Density-dependence in Single-species Populations of Plants

5.5 Maximum Sustainable Yield

Chapter 6

Populations in Space

Placeholder

6.1 Source-sink Dynamics

6.1.1 Complications and cases

6.2 Metapopulations

6.2.1 The classic Levins model

6.2.2 Propagule rain

6.2.3 The rescue effect and the core-satellite model

6.2.4 Parallels with Logistic Growth

6.2.5 Levins *vs.* Hanski

6.2.6 Habitat Destruction

6.2.7 Illustrating the effects of habitat destruction

6.2.8 Different assumptions lead to different predictions about rates of extinction

6.3 Hanski's incidence function

Chapter 7

Direct competition and mutualism

Placeholder

7.1 Interspecific competition

7.1.1 Equilibria

7.1.1.1 Isoclines

7.1.1.2 Finding equilibria

7.1.1.3 Coexistence — the invasion criterion

7.1.1.4 Other equilibria

7.1.2 Dynamics at the Equilibria

7.1.2.1 Determine the equilibria

7.1.2.2 Create the Jacobian matrix

7.1.2.3 Solve the Jacobian at an equilibrium

7.1.2.4 Use the Jacobian matrix

7.1.2.5 Three interesting equilibria

7.1.3 Return Time and the Effect of r

7.2 Interspecific mutualism

7.3 Generalized Lotka-Volterra models

7.4 Nonlinear direct competition

7.4.1 Investigate stability criteria

7.5 In fine

Chapter 8

Coexistence via competition, dispersal, and storage

Placeholder

8.1 Competition–colonization Tradeoff

8.1.0.1 What is a “site”?

8.1.0.2 Estimating colonization and mortality rates

8.1.1 Habitat destruction

8.1.1.1 Multispecies competition–colonization tradeoff and habitat destruction

8.2 This models S species with a competition–colonization tradeoff

8.3 Adding Reality: Finite Rates of Competitive Exclusion

8.3.1 Successional niche model

8.3.2 Dynamics of the successional niche model with a high rate of competitive exclusion

8.3.3 Dynamics of the successional niche model with a low rate of competitive exclusion

8.3.4 Dynamics of the successional niche model with an intermediate disturbance regime

8.3.5 Dynamics of the pure successional niche model

8.3.6 An analytical solution

8.3.7 Extinction debt with the successional niche?

8.4 Storage effect

8.5 functions for responses

8.6 differential responses

8.6.1 three graphs for two environments

8.7 An example of the temporal storage effect

8.7.1 Building a simulation of the storage effect

8.7.1.1 Simulating dynamics

8.7.1.2 Examining characteristics of the storage effect

Chapter 9

Consumer-resource Interactions

Placeholder

9.1 Ratio dependence

9.1.1 Dynamics of ratio dependent predation

9.2 Prey dependence

9.2.1 Dynamics of prey-dependent predation

9.3 Interlude: Functional response - what an individual predator does

9.4 Stability analysis for the prey-dependent Lotka–Volterra model

9.5 Resource growth functions

9.5.1 Consumption of a renewable resource

9.5.2 Prey carrying capacity and type II functional response

9.6 Paradox of enrichment

9.7 Consumption of two resources

9.7.1 Interpreting consumption and assimilation

9.8 In fine

Chapter 10

Host-parasitoid relations

Placeholder

10.1 Independent and random attacks

10.1.1 Simulating simple host-parasitoid dynamics

10.1.1.1 Simulating Random Attacks

10.2 Aggregation leads to coexistence

10.2.1 Equilibria for a discrete-time model

10.3 Dynamics of the May host–parasitoid model

10.4 *In Fine*

Chapter 11

Disease

Placeholder

11.1 Closed epidemics

11.1.1 Density-dependent transmission

11.1.2 Frequency-dependent transmission

11.2 Open epidemics

11.3 Modeling data from Bombay

11.3.1 Optimization

Chapter 12

Consumer-resource Competition and Mutualisms

Placeholder

12.1 Competition for one resource

12.2 Competition for Substitutable resources

12.3 Tilman's resource ratio with essential resources

12.4 Consumer-resource mutualism

12.4.1 A model of uni-directional mutualism

12.5 Plant-soil feedbacks

12.6 Simulations for learning

12.6.1 `lvg()`

12.6.2 Systematic simulations

12.6.3 Checking simulation results

12.6.4 Comparing simulation results to analytical solutions

12.6.5 One last thing: using the Jacobian matrix

12.6.6 *In Fine*

Chapter 13

Food webs and other networks

Placeholder

13.1 Food Web Characteristics

13.2 Does omnivory destabilize food chains?

13.2.1 Multi-species Lotka–Volterra notation

13.2.2 Background

13.2.3 Implementing Pimm and Lawton’s Methods

13.2.4 Shortening the Chain

13.2.5 Adding Omnivory

13.2.6 Comparing Chain A versus B

13.2.7 Re-evaluating Take-Home Messages

13.3 Perhaps mutualism stabilizes bipartite networks

13.3.1 Creating bipartite webs

13.3.2 Testing effects of nestedness and modularity on network resilience.

13.3.3 Modeling causal networks

13.4 Exploring Pandora, right here at home

13.4.1 Properties assessed

Chapter 14

Diversity

Placeholder

14.1 Background

14.2 Species Composition

14.2.1 Measures of abundance

14.2.2 Dissimilarities and Distances

14.2.3 Displaying multidimensional distances

14.3 Diversity

14.4 Species diversity

14.5 Rarefaction and total species richness

14.5.1 An example of rarefaction and total species richness

14.6 Distributions

14.6.1 Log-normal distribution

14.6.2 Other distributions

14.6.3 Pattern *vs.* process

14.7 Neutral Theory of Biodiversity and Biogeography

14.7.1 Different flavors of neutral communities

14.7.2 Investigating neutral communities

14.8 Species–area relations

14.8.1 Island biogeography

14.9 Diversity Partitioning

14.9.1 Partitioning species–area relations

14.10 In fine

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